



TECHNICAL MANUAL

My Health Passport

System Architecture, SLK-581 & SHA-256 Technical Reference,
and Researcher Integration Guide

Document version: 2.0

Prepared for: Research teams integrating with My
Health Passport

Applies to: index.html, portal.html, phone.html,
link_gen.html

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1. Executive Summary

What this document covers, and who it is for

My Health Passport is a small set of static, client-side web pages that let a research participant generate a pseudo-anonymised identifier — a **My Health Passport Number** (an SLK-581 statistical linkage key¹) or, optionally, a **My Health Passport Barcode** (a SHA-256 cryptographic hash² of that number) — entirely inside their own browser, and share only that identifier with a research team.

This manual is written for researchers, data managers, and HREC applicants who need to integrate My Health Passport into a REDCap or Qualtrics survey, a phone-based interview workflow, or an email-based study contact process. It supersedes all earlier revisions of this guide.

The system is deliberately simple: there is no server, no database, and no account system. Every calculation — building the SLK-581, hashing it, checking a form field — happens locally on the participant's device. The pages exist only to standardise how that identifier is generated and handed off to a survey platform, a phone, or an email client.

What has changed since the previous edition of this guide:

- The main generator page has been renamed and now sits at the root of the project as `index.html` (previously distributed as `SLK581_builder.html`).
- A dedicated **Portal Link Generator** (`link_gen.html`) is now available, removing the need to hand-build query strings for most researchers.
- `phone.html` now supports both an SMS reply workflow and a new **email reply workflow**.
- This edition adds a full technical briefing on SLK-581 and SHA-256, with cited references, for teams that need to justify the approach in an ethics submission or data governance review.

2. System Architecture & File Structure

Where everything lives, and what each file does

My Health Passport is a static site: four HTML files sitting in a single root directory, with no build step and no server-side component. All four files are self-contained — each can be opened directly in a browser or hosted from any plain web server or static host (e.g. GitHub Pages).

File	Role	Typical audience
<code>index.html</code>	The main public page. Explains the project, generates a Passport Number/Barcode directly, and links to the Technology, Privacy, and Resources sections.	Participants arriving directly; general public
<code>portal.html</code>	A focused landing page for survey-linked participants. Generates the identifier and, on consent, redirects to a survey URL with the identifier appended as a query parameter.	Participants following a REDCap/Qualtrics invitation link
<code>phone.html</code>	A landing page for phone- or email-based studies. Generates the identifier and, on consent, hands off to the participant's SMS or email app with the identifier pre-filled in a reply.	Participants contacted by phone interview or email
<code>link_gen.html</code>	An internal tool for researchers. Builds correctly-formatted <code>portal.html</code> / <code>phone.html</code> links (with QR code output) without needing to hand-construct query strings.	Researchers, study coordinators

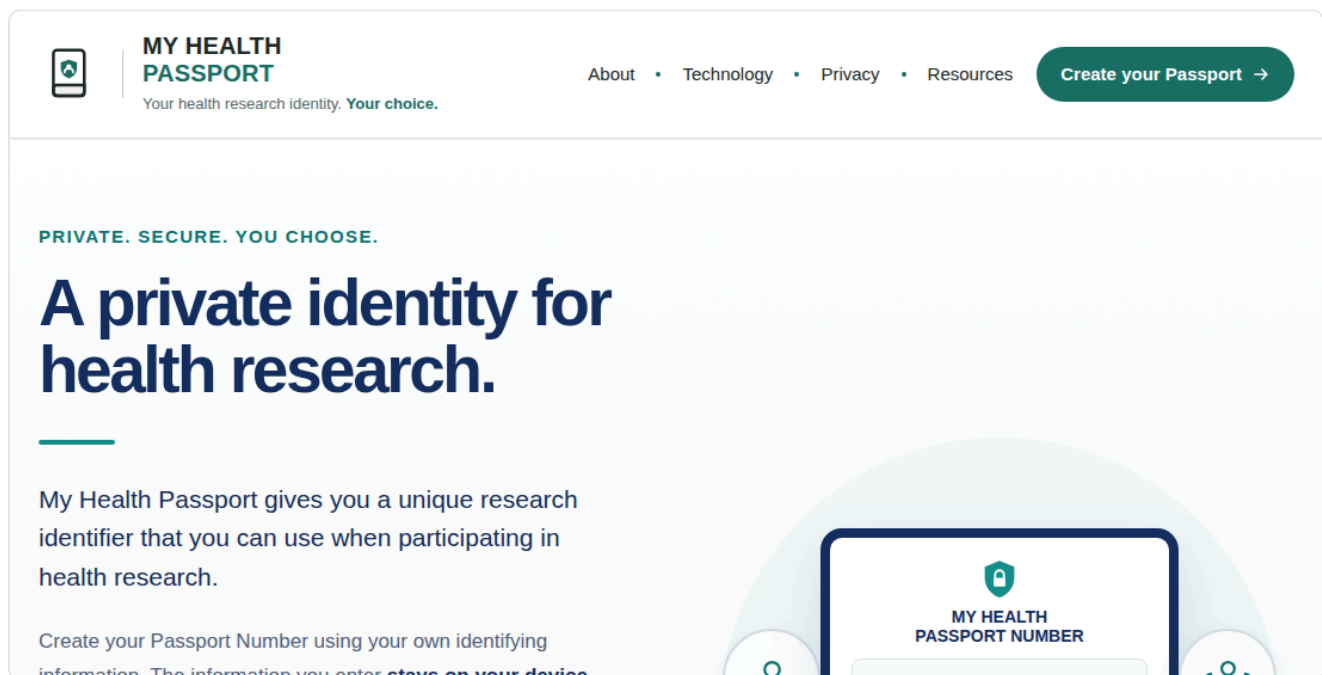
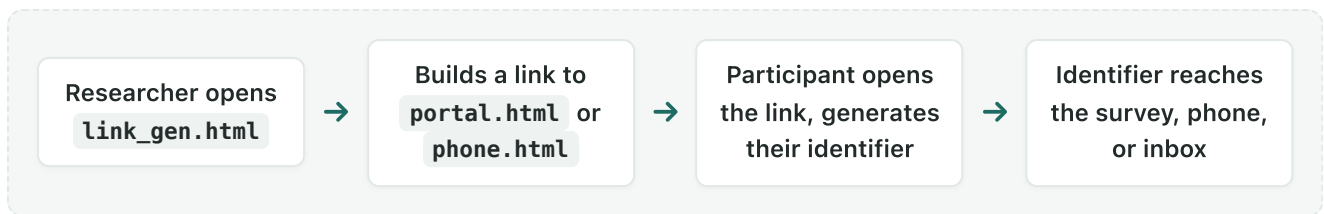


Figure 1 — `index.html`, the project's public entry point, showing the shared header and navigation used across all four pages.



Live deployment: this project is deployed at <https://my-health-passport.netlify.app/> . Because all four files live in the same directory, links between them are simple relative paths (e.g. `portal.html` , `phone.html`) — every query parameter described in this document works identically regardless of the base URL it's hosted under.

3. Technical Briefing: The SLK-581 Statistical Linkage Key

What it is, how it is built, and what it can and cannot do

3.1 Origin and purpose

The Statistical Linkage Key, 581-character format (SLK-581) is a data linkage standard developed and published by the Australian Institute of Health and Welfare (AIHW)³. It was created to solve a specific problem in health and community-service data collection: allowing records relating to the same person to be brought together across separate datasets, without those datasets ever exchanging a person's name, address, or other directly identifying information.

Rather than encrypting or storing a name, SLK-581 derives a short, largely non-reversible code from a small number of demographic elements the person already knows: parts of their family and given name, their date of birth, and their recorded sex. Two records created from the same underlying details will produce the same SLK-581, which is what allows linkage — without either dataset needing to see the other's raw identifying fields.

3.2 Construction algorithm

My Health Passport implements the SLK-581 algorithm exactly as specified by AIHW⁴. The 14-character key is assembled from four components, concatenated in order:

Component	Characters	Rule
Family name	3	The 2nd, 3rd and 5th letters of the family name, counting alphabetic characters only. A missing letter (e.g. a short surname) is filled with the digit <code>2</code> .
Given name	2	The 2nd and 3rd letters of the first given name, with the same fallback rule.
Date of birth	8	Recorded as <code>DDMMYYYY</code> .
Sex	1	A single digit: 1 (male), 2 (female), 3 (intersex/indeterminate), or 9 (not stated), consistent with the ABS/AIHW standard for recording sex.

Worked example

Family name `O'Brien-Smith` → alphabetic letters `OBRIENSMITH` → 2nd/3rd/5th letters → `BRE`

Given name `Elizabeth` → 2nd/3rd letters → `LI`

Date of birth `15 June 1990` → `15061990`

Sex: Female → `2`

Resulting SLK-581: `BRELI150619902`

This is exactly the calculation performed by the generator on `index.html`, `portal.html`, and `phone.html` — entirely in the participant's browser, using JavaScript equivalent to the reference implementation above.

3.3 Properties and limitations

- **It is not encryption.** SLK-581 is a deterministic transformation, not a cryptographic scheme. Given a plausible name, date of birth, and sex, the same key can in principle be reconstructed — SLK-581's privacy value comes from what it withholds (the full name, address, and other fields), not from mathematical irreversibility.
- **It is not guaranteed unique.** Two different people can share an SLK-581, particularly common names combined with common birth dates. Independent AIHW evaluation of SLK-based linkage against name-based linkage found strong but not perfect performance: a stepwise SLK-based strategy achieved a 99.7% positive predictive value and 98.5% sensitivity against a name-based reference standard in one aged-care linkage study⁵, and AIHW has separately reported that a small proportion of residents in large administrative datasets share an SLK-581 with at least one other person⁶.
- **It is widely used, not experimental.** SLK-581 underpins linkage in numerous Australian health and community-service data collections and has been the subject of peer-reviewed evaluation in registry-based research⁷.
- **Governance still matters.** AIHW and equivalent bodies recommend SLK-581 be used within an appropriately governed linkage project — with ethics approval, and additional identifying variables where higher linkage confidence is required — rather than as a stand-alone guarantee of identity.

Do not present SLK-581 as anonymisation. It is more accurately described as *pseudonymisation*: identifying detail is withheld from the receiving dataset, but the key is derived from, and in some circumstances could be linked back to, identifying information the participant already knows. HREC submissions should describe it in these terms — see Section 12.

4. Technical Briefing: The SHA-256 Cryptographic Hash

The optional "My Health Passport Barcode"

4.1 What SHA-256 is

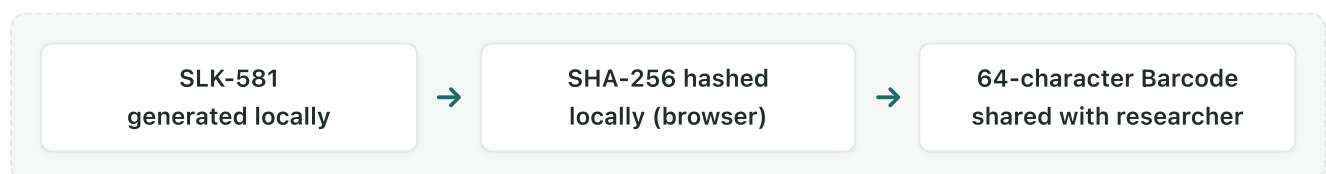
SHA-256 ("Secure Hash Algorithm, 256-bit") is a member of the SHA-2 family of cryptographic hash functions, specified by the U.S. National Institute of Standards and Technology (NIST) in Federal Information Processing Standard 180-4⁸. A hash function takes an input of any length and produces a fixed-length output — for SHA-256, always 256 bits, conventionally written as 64 hexadecimal characters.

Three properties matter for this project:

- **Deterministic:** the same input always produces the same 64-character output, which is what allows the same Passport Number to always generate the same Barcode.
- **One-way:** there is no practical method to recover the original input from the hash output alone. This is a fundamentally different property to SLK-581, which withholds detail but does not cryptographically destroy it.
- **Avalanche effect:** changing a single character of the input (e.g. one letter of a name) produces a completely different, unrelated-looking 64-character output, which is why a Passport Barcode gives no partial information about the Passport Number it came from.

4.2 Why it is used here (the "My Health Passport Barcode")

When barcode mode is enabled, the portal computes `SHA-256(Passport Number)` locally in the participant's browser, using the standard Web Cryptography API (`crypto.subtle.digest`), and displays and shares that hash — the **My Health Passport Barcode** — instead of the raw SLK-581. This gives studies a way to obtain a stable, unique-per-input research identifier while adding a one-way transformation on top of SLK-581, for projects that want that additional layer.



Because the hash is one-way, reversing it to recover the original SLK-581 — and therefore the demographic detail behind it — is not practically feasible. It is important, however, to describe this accurately rather than in absolute terms: cryptographic best practice is to describe SHA-256 as making reversal *computationally infeasible* with current methods, not mathematically impossible for all time.

4.3 Standards and approvals

SHA-256 is standardised by NIST under FIPS 180-4⁸ and is one of the hashing algorithms approved for general-purpose use by the Australian Signals Directorate (ASD) under the Australian Government

Information Security Manual (ISM)⁹. Current ASD guidance approves SHA-2 family algorithms (including SHA-256) for general use, while recommending that new systems intended for long-term use prefer the longer SHA-384 or SHA-512 variants, noting that ASD's approval of SHA-224 and SHA-256 specifically is not intended to extend beyond 2030 for interoperability planning reasons. This is a reasonable, currently-approved choice for this project's purpose; it is not a claim that SHA-256 is invulnerable indefinitely.

5. Portal Workflow Overview

How a participant experiences the process end to end

1. **Participant landing:** the participant opens a link to `index.html` , `portal.html` , or `phone.html` , depending on how they were contacted.
2. **Identity key generation:** the participant enters their family name, given name, date of birth, and sex. The page builds their SLK-581 (and, if enabled, its SHA-256 Barcode) live, showing a progress indicator as each field is completed.
3. **Consent to share:** the participant reviews the generated identifier and taps the clearly labelled share action — "Share your My Health Passport Number/Barcode" (survey), or the equivalent SMS/Email reply action.
4. **Hand-off:** depending on the page, this either opens the destination survey in a new tab with the identifier appended as a URL parameter, or opens the participant's own SMS or email app with a pre-filled reply.

The screenshot displays a web form for generating a Health Passport. On the left, the input fields are filled: Family name 'OBrien-Smith', First given name 'Elizabeth', Date of birth (Day '15', Month 'June', Year '1990'), and Sex 'Female (2)'. Each input field has a red asterisk and a note '* This field is compulsory'. On the right, the generated 'Your Health Passport Number' is 'BRELI150619902', displayed in individual boxes. Below it, the 'Your Health Passport Barcode' is shown as a barcode and a long alphanumeric string: 'D53AD7FD482AAE1398856386EC827484CB8E89CA8105F8DD2B5EEBC2393F8EC6'. A green box at the bottom states: 'Nothing is sent anywhere. Every calculation happens in your browser — the details you enter never leave this page or get stored.'

Figure 2 — The generator on `index.html` , showing a completed Passport Number and its optional SHA-256 Barcode.

At no point does any of the entered detail (name, date of birth, sex) leave the participant's device. Only the finished identifier — the SLK-581 or its SHA-256 Barcode — is ever transmitted, and only once the participant actively chooses to share it.

`portal.html` presents the same generator inside a three-step flow, with each step lighting up as the participant progresses:



MY HEALTH PASSPORT PORTAL

Unlock Your Survey

1

Enter Your Details ⓘ



OBrien-Smith, Elizabeth · 15 Jun 1990 · Female

 Edit

2

Get your Unique ID



Your Anonymised My Health Passport Number

BRELI150619902



3

Share your My Health
Passport Number



Selected letters from your first and last names, as well as your date of birth and encoded gender, are used to generate your unique identifier.

You consent to share your unique identifier with the researchers conducting the survey. Your first name or last name in whole will not be shared.

Powered by



MY HEALTH
PASSPORT

Compatible with



REDCap



qualtrics

Your health research identity. **Your choice.**

[Visit Main My Health Passport Website →](#)



Figure 3 — `portal.html` 's step-by-step flow: enter details, generate the ID, then share it with the survey.

6. Using the Portal Link Generator (link_gen.html)

The recommended way to build a distribution link

Rather than hand-constructing query strings (Section 7), most researchers should use `link_gen.html`. It asks three questions and builds the correct link automatically:

1. **Link type** — Online Survey, SMS phone message, or Email.
2. **Contact detail** — the survey name and URL, the reply phone number, or the reply email address, depending on the link type chosen.
3. **Identifier type** — My Health Passport Number (SLK-581) or My Health Passport Barcode (SHA-256).



FOR RESEARCHERS

Portal Link Generator

Create specialised links for your research project. These links send participants to a landing page where they can confirm or generate their My Health Passport Number or Barcode, supporting secure identity linkage for web surveys, SMS invitations, or email-based study contact — no first or last names collected. Turn any link into a QR code or short URL using your usual tools.

LINK TYPE

I want to create a link for:

- ☒ Online Survey (e.g. REDCap or Qualtrics)
- ☐ An SMS phone message
- ☐ An Email

Enter the name of your survey

ED Injury Study

Enter your survey's URL

<https://redcap.rmh.org.au/surveys/?s=8X9Y7Z> ✓

IDENTIFIER TYPE

Figure 4 — `link_gen.html`, configured to build a survey link for an "ED Injury Study".

The tool live-validates a survey URL (it must begin with `https://`), generates the finished link in a copyable field, and can render that link as a scannable QR code for printed materials — useful for posters, flyers, or waiting-room signage.

Where the QR code comes from: the QR code image is produced by a third-party rendering service, which receives only the finished portal link itself (never any participant data, since no participant has interacted with the link yet). Study teams with strict data-egress requirements should be aware that this one step involves an external request, unlike every other part of this system.

7. Constructing Links Manually (Advanced)

For teams integrating link generation into their own tooling

Both `portal.html` and `phone.html` read plain URL query parameters. This section documents them directly, for teams who want to generate links programmatically (e.g. from a REDCap API or a mail-merge) rather than via `link_gen.html`.

portal.html (survey links)

```
https://my-health-passport.netlify.app/portal.html?siteTitle=[SURVEY_NAME]&siteUrl=[SURVEY_URL]&barcode=[true|false]
```

Parameter	Required	Notes
<code>siteTitle</code>	Yes	Shown in the Share button label. Spaces must be URL-encoded as <code>%20</code> .
<code>siteUrl</code>	Yes	The destination survey URL. The identifier is appended to this as <code>?slk=</code> or <code>&barcode=</code> .
<code>barcode</code>	No (default <code>false</code>)	<code>true</code> shares the SHA-256 Barcode instead of the raw SLK-581.

phone.html (SMS and email links)

```
https://my-health-passport.netlify.app/phone.html?SurveyPhone=[PHONE]&barcode=[true|false]&linkType=sms
https://my-health-passport.netlify.app/phone.html?replyemail=[EMAIL]&barcode=[true|false]&linkType=email
```

Parameter	Required	Notes
<code>SurveyPhone</code>	For SMS links	The number the participant's reply SMS should be addressed to.
<code>replyemail</code>	For email links	The address the participant's reply email should be addressed to.
<code>linkType</code>	Recommended	<code>sms</code> or <code>email</code> . Tells the page which reply channel to present. If omitted, the page infers the channel from whichever of <code>SurveyPhone</code> / <code>replyemail</code> is present.
<code>barcode</code>	No (default <code>false</code>)	As above — when <code>true</code> , the SMS or email body contains the SHA-256 Barcode instead of the raw SLK-581.

Example — REDCap survey, Barcode mode enabled

```
https://my-health-passport.netlify.app/portal.html?siteTitle=ED%20Injury%20Study&siteUrl=https://redcap.rmh.org.au/surveys/?s=8X9Y7Z&barcode=true
```

Example — phone interview, SMS reply

<https://my-health-passport.netlify.app/phone.html?SurveyPhone=%2B61400000000&barcode=false&linkType=sms>

Example — email-contacted participant

<https://my-health-passport.netlify.app/phone.html?replyemail=study%40example.org&barcode=false&linkType=email>

8. Survey Platform Setup (REDCap & Qualtrics)

Capturing the identifier once it reaches your survey

Critical requirement for multi-page surveys: the receiving field **must be on the very first page** of the survey. If it sits on page 2 or later, page navigation strips the incoming URL parameter before it can be captured.

REDCap setup

1. Open the project's **Online Designer**.
2. On the **first survey instrument**, add a new Text Box field.
3. Set the Field Name to exactly `slk` — or `barcode` if the link uses Barcode mode.
4. Under Action Tags, add `@READONLY` to prevent participant editing.

Qualtrics setup

1. Open the **Survey Flow** tab.
2. Add a new **Embedded Data** element.
3. Name the field exactly `slk` — or `barcode` if the link uses Barcode mode — and leave the value blank.
4. Drag the Embedded Data block to the very top of the Survey Flow, above Question Block 1.

9. SHA-256 Encrypted Barcode Mode

Enabling and receiving the hashed variant

Barcode mode (`barcode=true`) is described technically in Section 4. Operationally, enabling it changes three things simultaneously, on both `portal.html` and `phone.html` :

- The participant sees the "My Health Passport Barcode" panel instead of the raw Passport Number panel.
- The Share button label and consent disclaimer update to describe sharing a Barcode rather than a Passport Number.
- The value transmitted — whether by URL parameter, SMS, or email — is the 64-character SHA-256 hash, never the underlying SLK-581.



MY HEALTH PASSPORT PORTAL

Generate your secure Phone ID

1

Enter Your Details ⓘ



OBrien-Smith, Elizabeth · 15 Jun 1990 · Female

 Edit

2

Get your Unique ID



Your Anonymised My Health Passport Barcode



D53AD7FD482AAE1398856386EC827484CB8E89CA81
05F8DD2B5EEBC2393F8EC6



Help reading my ID aloud



3

Reply via SMS with your
My Health Passport
Barcode



Selected letters from your first and last names, as well as your date of birth and encoded gender, are used to generate your unique identifier.

You consent to share your unique identifier with the researchers conducting the survey. Your personal identifying data has been irreversibly encrypted.

Powered by



MY HEALTH
PASSPORT

Compatible with



REDCap



qualtrics

Your health research identity. **Your choice.**

[Visit Main My Health Passport Website](#)

Figure 5 — `phone.html` in Barcode mode: the SLK-581 panel is replaced entirely by the SHA-256 Barcode panel.

Field name differs from standard mode. When a link uses Barcode mode, the receiving field on the survey side must be named `barcode`, not `slk` (see Section 8). The two modes are mutually exclusive per link — build and label only the field matching the mode your distribution link actually uses.

10. Mobile Phone / SMS Reply Workflow

For phone interviews and other non-web-survey studies

For phone-based interviews, `phone.html` lets a participant generate their identifier on their own device and reply directly by SMS — useful for establishing an anonymous, persistent identifier at the very start of an interview, before any interview content is discussed.

1. **Send the link:** at the start of the call, the interviewer sends the `phone.html` link (built via `link_generator.html`, see Section 6) to the participant by SMS.
2. **Participant generates their ID:** the participant taps the link and completes the form on their own device.
3. **Participant replies:** tapping the Share action opens the participant's own messaging app, pre-filled with a message addressed to the study's reply number, containing the generated identifier.
4. **Researcher logs the ID:** once sent, the interviewer receives the SMS and records the identifier against the interview record.

Verbal fallback: `phone.html` also renders each character of the Passport Number using the NATO phonetic alphabet (e.g. B → "Bravo"), so an interviewer can capture it verbally during the call if a written record is needed immediately, without waiting for the SMS to arrive.

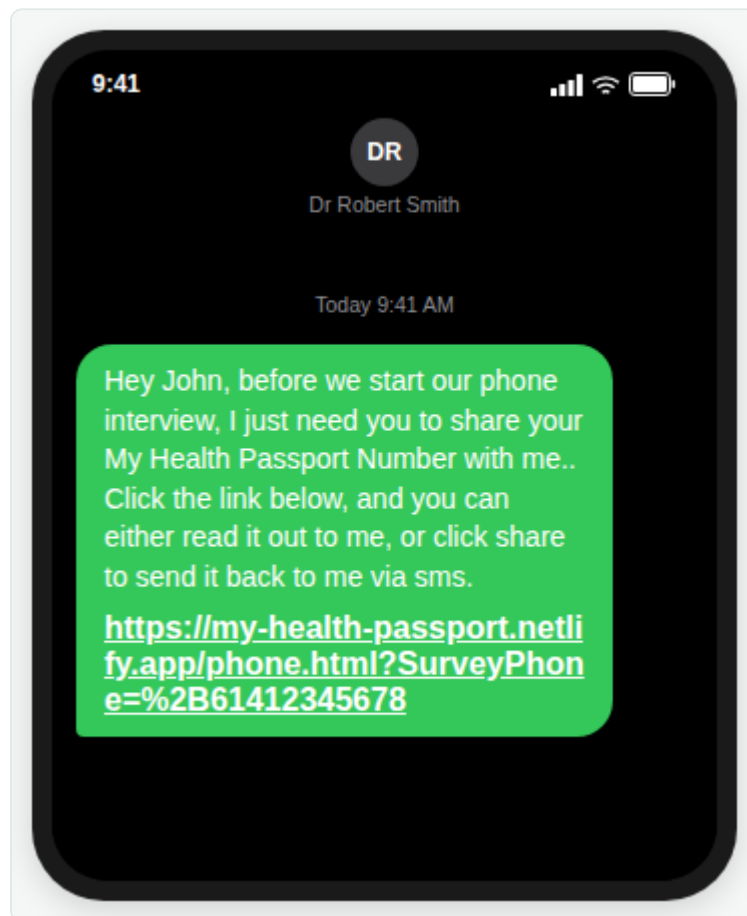


Figure 6 — Example SMS invitation, with the `phone.html` link addressed back to the interviewer's own number via `SurveyPhone`.

Delivery is participant-initiated. The Share action opens a pre-filled message in the participant's own app — it does not send automatically. The participant must still tap Send. Interviewers should confirm receipt before proceeding, particularly in time-sensitive settings.

11. Email Reply Workflow

New: for participants contacted by email

`phone.html` also supports an email-based reply, for studies that make first contact by email rather than phone. The mechanism mirrors the SMS workflow in Section 10, substituting a `mailto:` hand-off for an `sms:` hand-off.

1. Build an email-type link in `link_gen.html`, supplying the study's reply email address.
2. Send that link to the participant by email, as part of the standard study invitation.
3. The participant opens the link, completes the form, and taps **"Reply via Email with your My Health Passport Number"** (or Barcode, if enabled).
4. Their default mail client opens with a new message addressed to the study's reply address, subject and body pre-filled with the identifier. The participant reviews and sends it.

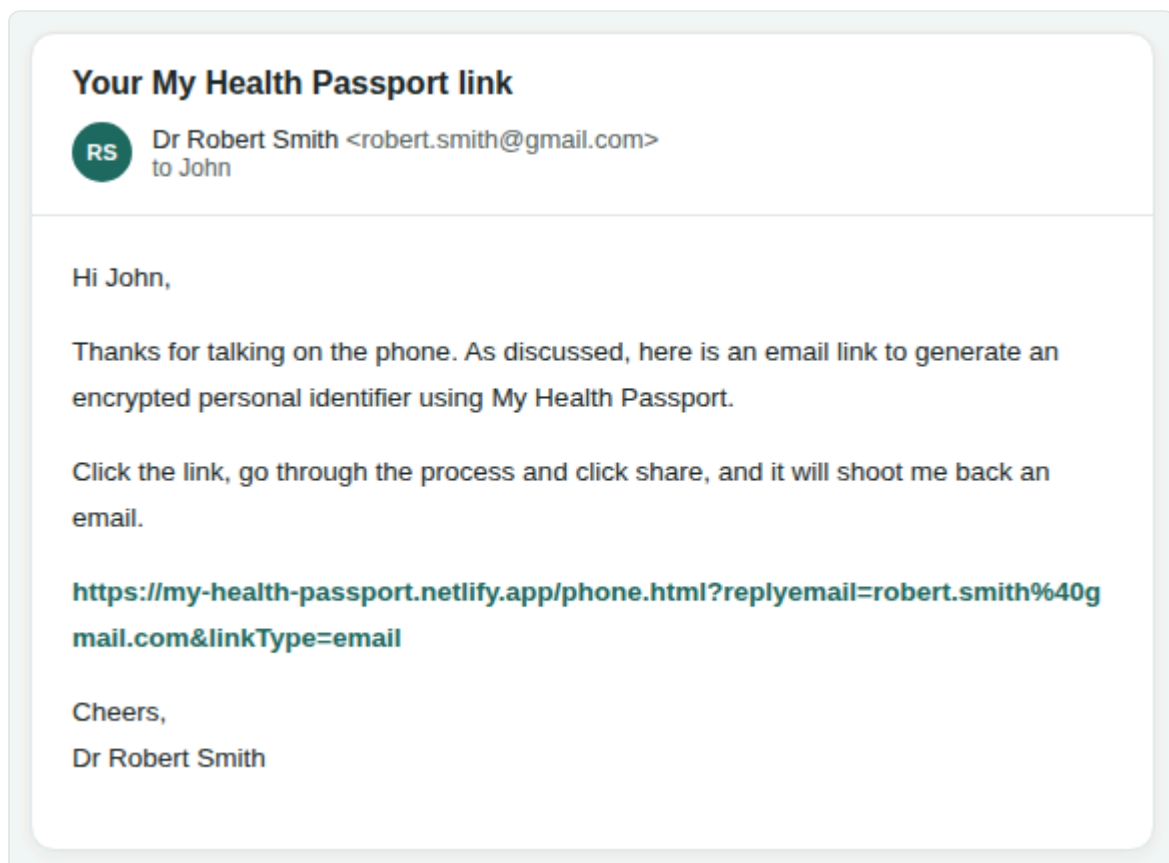


Figure 7 — Example email invitation, with the `phone.html` link addressed back to the researcher's email via `replyemail` and `linkType=email`.

Which reply channel is shown is controlled by the `linkType` parameter (`sms` or `email`) described in Section 7; the page falls back to inferring the channel from whichever of `SurveyPhone` or `replyemail` is present if `linkType` is omitted, for compatibility with earlier links.

Consistent with the rest of the system: as with SMS, no identifying detail is transmitted automatically — only the finished identifier, and only once the participant reviews and sends the pre-filled email themselves.

12. HREC Ethics Statement Template

Suggested wording for ethics submissions

Researchers may adapt the following text for Human Research Ethics Committee (HREC) submissions. It has been updated to reflect the email reply channel and to describe SLK-581 as pseudonymisation rather than anonymisation, consistent with Section 3.3.

"Data collection will utilise the My Health Passport tools as a secure participant landing page. Participants will generate their Statistical Linkage Key (SLK-581) on their own device; no first or last names, addresses, or other directly identifying fields are captured or transmitted at any point. Upon explicit participant authorisation, only the resulting pseudonymised SLK-581 code — or, where the study elects to use the optional SHA-256 hashed Barcode mode, a one-way cryptographic hash derived from it — is transmitted into the research record, via URL parameter injection into the survey platform, or via a participant-initiated SMS or email reply. SLK-581 pseudonymises rather than anonymises participant data: while no directly identifying fields are transmitted, the key is derived from identifying information the participant already holds, and record linkage carries a small residual risk of collision between participants with similar names and birth dates, as documented in published AIHW evaluations."

13. Data Field Validation

Sense-checking entered details before they become part of a key

Because the SLK-581 and its SHA-256 Barcode are derived directly from whatever a participant types, the quality of the generated identifier is only as good as the quality of the input. `index.html`, `portal.html`, and `phone.html` all now run the same client-side validation on the two fields most prone to low-quality entry — date of birth and first name — before a Passport Number or Barcode is generated.

13.1 Date of birth sense-checking

The date of birth fields are checked against two rules the moment all three components (day, month, year) are entered. Either rule failing blocks key generation entirely — the participant sees an inline message and no Passport Number or Barcode is produced until the date is corrected.

Rule	Behaviour
No future dates	A date of birth later than today's date is rejected with "Date of birth cannot be in the future."
Maximum age of 110 years	A date of birth implying an age over 110 is rejected with "Please check the date of birth — age cannot exceed 110 years."

A third, implicit check comes as a natural consequence of how the first two are implemented: an impossible calendar date (for example, 31 February) is also rejected, with "Please enter a valid calendar date." — since browsers would otherwise silently roll an invalid date over into a different, incorrect real date instead of flagging it.

Where "unknown" date components are used: `index.html` still supports marking the day, month, or year of birth as unknown (recorded as `99 / 9999` per the SLK-581 standard — see Section 3.2). Validation is skipped entirely whenever any date component is marked unknown, since a partial date cannot be meaningfully checked against these rules.

13.2 Hypocorism (nickname) screening

Separately, the given-name field is checked against a maintained, proprietary reference list of over 140 common English-language hypocorisms — informal, shortened, or pet forms of a formal given name (for example, "Sam" for "Samuel," "Liz" for "Elizabeth," "Charlie" for "Charles"). The purpose is to gently prompt participants toward using the name recorded on their formal identity documents, since AIHW guidance on SLK-581 construction assumes formal given names are used as input.

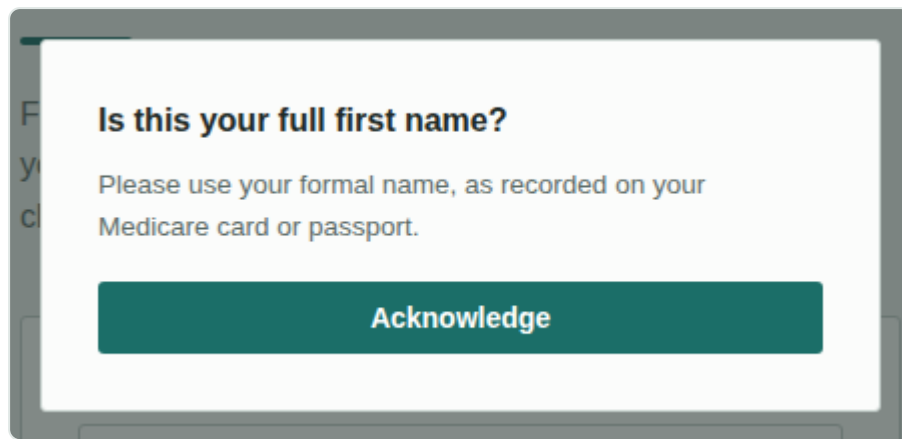


Figure 8 — The hypocorism prompt, shown after a participant enters a recognised nickname and moves to the next field.

How the check runs

1. On page load, the current reference list is fetched once from a dedicated Supabase table and held in memory as a simple lookup set.
2. When the participant finishes typing their given name and moves to the next field, the entered value is compared — case-insensitively — against that in-memory list.
3. If it matches a known hypocorism, a dialog appears: *"Please use your formal name, as recorded on your Medicare card or passport."*
4. The dialog has a single action, **Acknowledge**, which simply closes it. Once dismissed for a given entry, it will not reappear unless the participant changes the field to something else and back.

This is a prompt, not a restriction. Entering a hypocorism is not prohibited at any point. The dialog does not alter, clear, or reject the entered text, and the participant may proceed to generate their Passport Number or Barcode using the nickname exactly as typed if they choose to. The check exists purely to encourage, not enforce, higher-fidelity input.

Where the comparison happens

The entered name is never transmitted anywhere as part of this check. The reference list is fetched generically — the same list, for every visitor, before any name has been typed — and every subsequent comparison against what a participant enters happens entirely in the browser, using a local, in-memory lookup. No network request is made at the moment of comparison itself.

Fault-tolerant by design: if the reference list fails to load for any reason (offline use, an ad blocker, or the underlying table being temporarily unavailable), the check simply never triggers. It cannot block or interfere with Passport Number generation, which depends only on the date-of-birth and required-field checks described above.

14. Researcher Tools Suite

A complete outline of every tool provided for researchers and data managers

Beyond the participant-facing pages, four dedicated tools exist to support researchers and database administrators across the full lifecycle of an SLK-581-based project — from building keys in bulk from an existing dataset, to generating participant-facing links, to converting existing keys to hashed form, to upgrading a legacy database. All four are gathered on a single entry point, `researchers.html`, alongside a link to this manual.

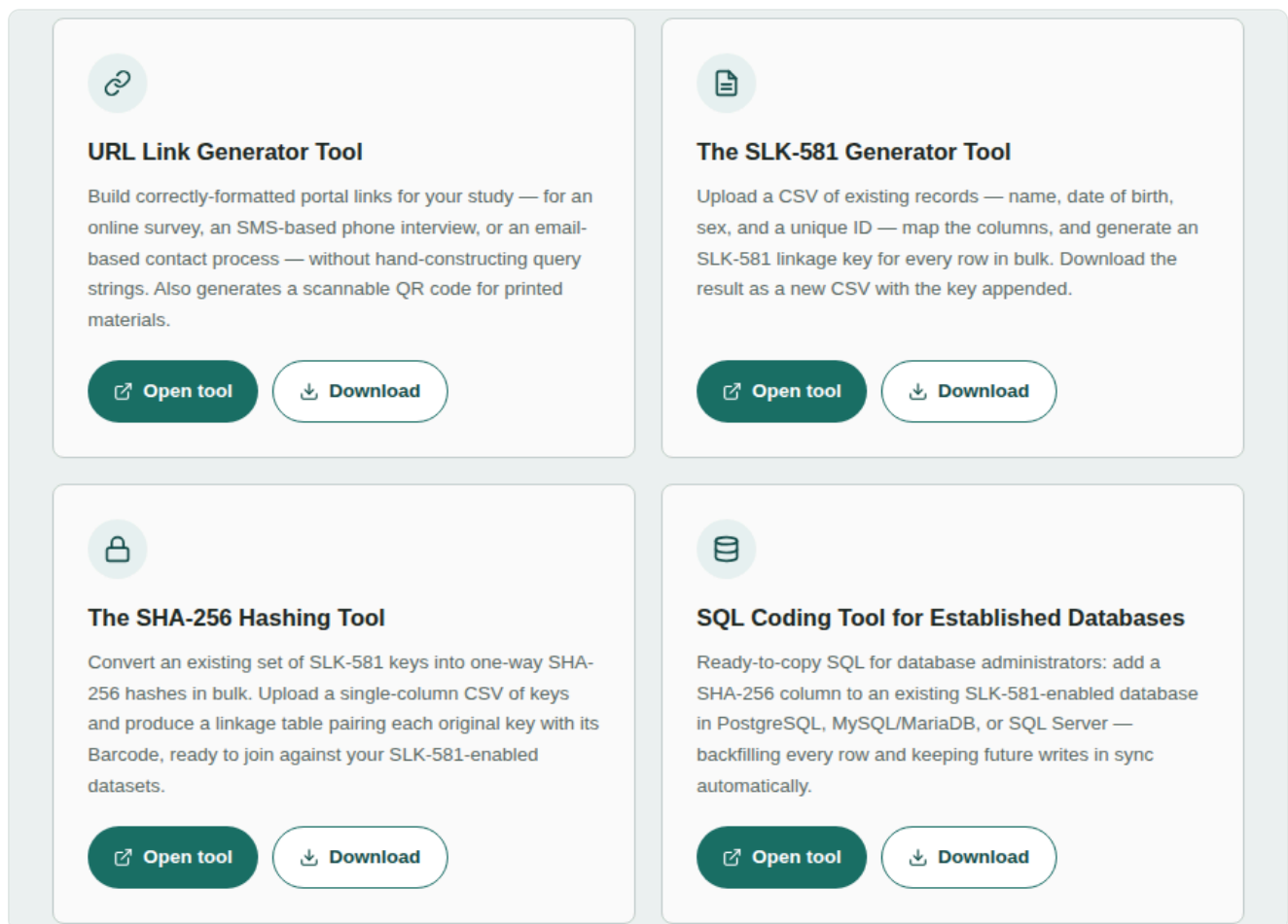


Figure 9 — The four researcher tool cards on `researchers.html`. Each opens the live tool in a new tab, or downloads the file to run offline.

14.1 URL Link Generator Tool (`link_gen.html`)

Builds correctly-formatted `portal.html` and `phone.html` links — for online surveys, SMS-based phone interviews, or email-based contact — without hand-constructing query strings, and renders the finished link as a scannable QR code. Fully documented in Section 6.

14.2 The SLK-581 Generator Tool (`csv-batch-generator.html`)

Converts an existing CSV of records — name, date of birth, sex, and a unique ID — into a new file with an SLK-581 key generated for every row, using exactly the same construction rules as the rest of the

toolset (Section 3.2).

Workflow

1. **Open:** the file is validated to contain at least 5 columns before anything else unlocks.
2. **Preview:** the first 5 data rows are shown across every uploaded column.
3. **Map columns:** five dropdowns — Unique ID, First Name, Last Name, DOB, Gender — are populated from the file's header row and matched to the researcher's own column names. A date-of-birth format selector (DD/MM/YYYY, YYYY-MM-DD, or MM/DD/YYYY) sits alongside, since a CSV cannot self-declare which convention it uses.
4. **Generate & download:** produces a new two-column CSV — the mapped Unique ID column, and the generated `SLK-581` column — leaving every other original column out of the output.

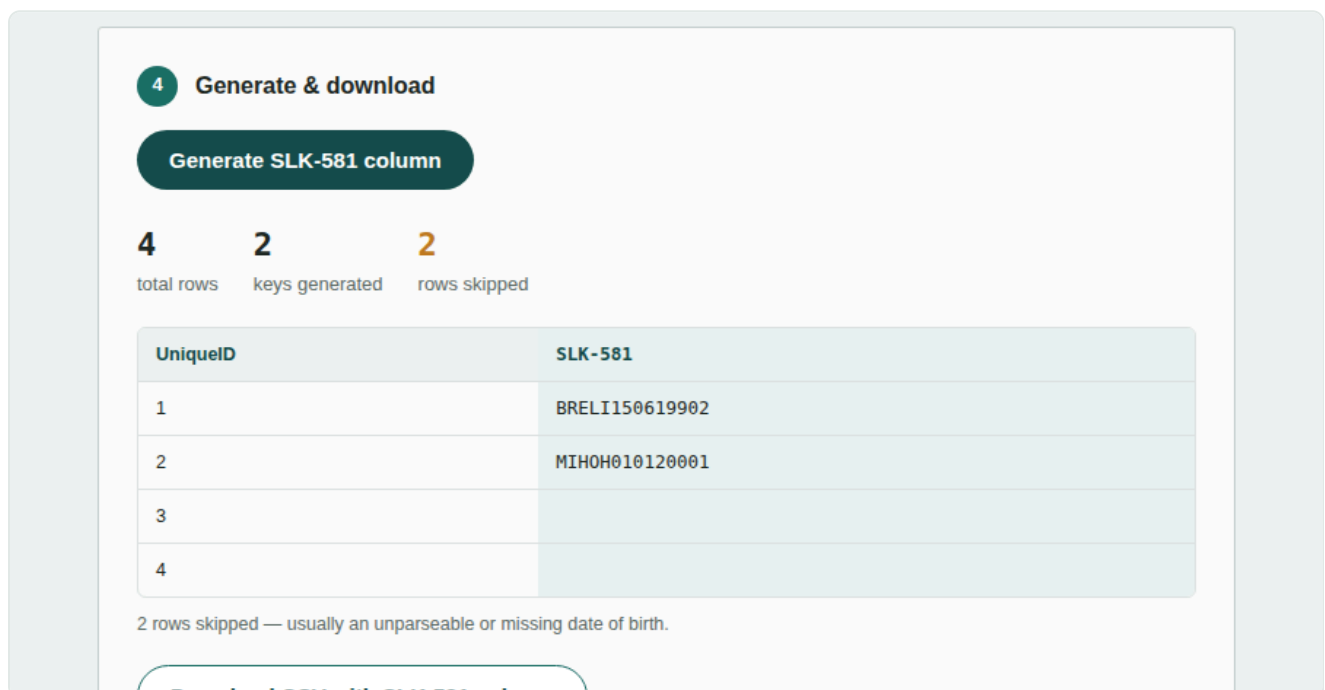


Figure 10 — Step 4 of the SLK-581 Generator Tool: generation statistics and a preview of the two-column output.

Rows that cannot be processed — an unparseable or missing date of birth, or a missing name — are left blank in the output rather than halting the batch; the total, generated, and skipped counts are shown after every run. Gender values are matched using the same case-insensitive coding described in Section 13.2.

Runs off the main thread. Parsing and key generation both execute inside a Web Worker, so the page never freezes on a large file. Tested directly against a 100,000-row file: parsed in under 0.3 seconds and generated in under 0.5 seconds, with a live progress indicator shown throughout.

14.3 The SHA-256 Hashing Tool ([csv-barcode-tool.html](#))

The bulk counterpart to Barcode mode (Section 9): converts an existing list of SLK-581 keys into their one-way SHA-256 hashes, producing a linkage table that pairs each original key with its Barcode.

Requirement	Detail
File shape	Exactly one column, one SLK-581 per row
Recommended scale	Under 250,000 rows — everything is held in browser memory rather than processed server-side in chunks
Value format	Letters and numbers only, with a mix of both — consistent with a genuine SLK-581

A file that doesn't meet these expectations still opens — a one-time dialog summarises what looks off (extra columns, symbols, or values that don't mix letters and numbers), and the researcher decides whether to continue. Before hashing, each value is uppercased and stripped of spaces, matching the normalisation used by the SQL approach in Section 14.4.

4 Save the result

slk_list-barcoded.csv **Save as...**

Exports the full file, including the Barcode column, as a new CSV. Edit the file name above before saving if you'd like.

SLK581	Barcode
BRELI150619902	d53ad7fd482aae1398856386ec827484cb8e89ca8105f8dd2b5eebc2393f8ec6
MIHOH010120001	d5086985646ab8d78d7ad4f91d1ee13b8076f402b0aa9f97e5bf8fb6222b6774

Figure 11 — Step 4 of the SHA-256 Hashing Tool: the finished linkage table, ready to save.

Also Worker-based. Parsing, validation, and hashing all run off the main thread. Tested against 100,000 keys: parsed in 0.15 seconds and hashed in 1.2 seconds, with the page remaining fully responsive throughout.

14.4 SQL Coding Tool for Established Databases ([research-upgradingDB.html](#))

Not an interactive tool but a reference document: ready-to-copy SQL for database administrators who want to add SHA-256 hashing directly inside an existing SLK-581-enabled database, rather than exporting data through a browser-based tool. Provides tested statements for PostgreSQL, MySQL/MariaDB, and SQL Server.

In every case, the SHA256 field is added as a *generated column* — calculated by the database itself, backfilling every existing row and recalculating automatically on every future insert or update, in a single statement with no separate trigger or backfill step. As with the browser-based hashing tool, the source value is capitalised and stripped of spaces before hashing.

Assumes a column named `SLK581` . Administrators using a different column name need to substitute it before running the supplied statements, along with their own table name.

14.5 The Researcher Landing Page (researchers.html)

The single entry point gathering all four tools above: an introduction to SLK-581 and SHA-256 for a researcher audience, a card for each tool with an Open action (launches the live tool in a new tab) and a Download action (saves the file locally via the browser's native download attribute), the Mind the Gap explainer reproduced in full, and a link through to this manual.

Appendix A — File & URL Quick Reference

Live base URL: `https://my-health-passport.netlify.app/`

File	Purpose	Key parameters
<code>index.html</code>	Main public page and stand-alone generator	—
<code>portal.html</code>	Survey-linked participant landing page	<code>siteTitle</code> , <code>siteUrl</code> , <code>barcode</code>
<code>phone.html</code>	Phone/email-linked participant landing page	<code>SurveyPhone</code> , <code>replyemail</code> , <code>linkType</code> , <code>barcode</code>
<code>link_gen.html</code>	Researcher tool for building the above links, with QR output	—
<code>csv-batch-generator.html</code>	Bulk SLK-581 generation from an uploaded CSV	—
<code>csv-barcode-tool.html</code>	Bulk SHA-256 conversion of an existing SLK-581 list	—
<code>research-upgradingDB.html</code>	Reference SQL for adding SHA-256 to an existing database	—
<code>researchers.html</code>	Single entry point linking all researcher tools and this guide	—

Survey field name	When to use it
<code>slk</code>	Standard mode (raw SLK-581 Passport Number)
<code>barcode</code>	Barcode mode (SHA-256 hash of the Passport Number)

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